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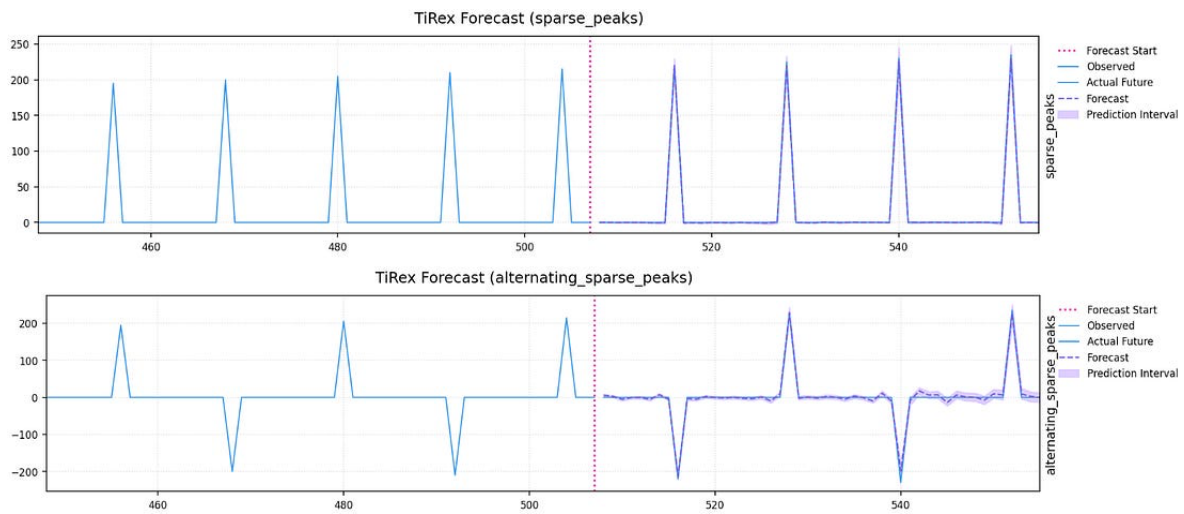
Forecast Sparse Data Better: TiRex vs Traditional Models - Practical Tutorial

An xLSTM based foundation model for zero-shot forecasting

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TiRex forecasts on sparse time series with prediction length=48 (Image by author)

If you follow recent research, you've likely seen headlines about **xLSTM**.

xLSTM[1] is a modern upgrade to the classic LSTM architecture, with strong results in computer vision[2], image segmentation[3], and related tasks.

The research team behind it pushed further — testing xLSTM for **zero-shot time-series forecasting**. The result is **TiRex[4]**, a foundation model built for time-series tasks. Key characteristics of TiRex:

- **Pretrained on diverse data:** Chronos, GIFT-Eval, and synthetic datasets.
- **Compact:** Contains just **35M parameters**, far smaller than models like Chronos-Bolt-Base (200M), Toto (151M), and TimesFM-2.0 (500M).
- **Superior Performance:** Currently ranks **#1 on the GIFT-Eval benchmark**, surpassing **Toto** within a month. The competition is tight — a sign of rapid progress in this space!

- **Probabilistic forecasting:** Predicts **9 quantiles**, enabling decision-aware forecasting.

While testing TiRex, I found it performs unexpectedly well on **intermittent data** — a challenging area for many models.

In this article, we'll go through 2 tutorials:

1. Compare **TiRex** with the recent foundation model **Timer-XL** on synthetic sparse data.
2. Benchmark **TiRex** against statistical models (IMAPA, Croston, etc.) on 2 BOOM datasets — featuring sparse and irregularly sampled series. As usual, we apply rigorous rolling origin evaluation on the full test set.

Let's get started!



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Part 1 - TiRex vs Timer-XL on Synthetic Sparse Data

In this section, we benchmark 2 models on a simple sparse-data task — part of the same setup I used when testing [DeepSeek](#).

The goal is to assess whether foundation models can handle intermittent data, where only a few specialized models perform well without manual tuning...

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